

ABSTRACT

The **ScanPay: A Mobile-Based Scan-and-Pay Retail Billing System for Queue-Free Shopping** is a mobile application designed to automate and streamline the retail shopping and billing process in supermarkets and shopping malls. Traditional billing systems often rely on manual checkout counters and long waiting queues, which can lead to delays, billing errors, and inefficient customer service. The application incorporates barcode and QR-code-based product scanning that allows customers to scan items using their mobile devices, add products to a digital cart, and complete payments through secure online payment gateways. It ensures accurate bill generation and proper maintenance of transaction records, supporting efficient retail management and secure digital payments. By digitizing core shopping and billing operations, the system improves accuracy, reduces manual effort, minimizes checkout time, and enhances customer convenience. This project is aligned with the United Nations Sustainable Development Goals, including **SDG 8: Decent Work and Economic Growth** to improve retail business efficiency, **SDG 9: Industry, Innovation and Infrastructure** by implementing smart retail technology solutions, and **SDG 12: Responsible Consumption and Production** which ensures transparent digital billing and efficient resource management.

LIST OF ABBREVIATIONS

ACRONYM	FULL FORM
SDLC	Software Development Life Cycle
QR	Quick Response
UPI	Unified Payments Interface
API	Application Programming Interface
JWT	JSON Web Token
DBMS	Database Management System
UI	User Interface
UX	User Experience
PDF	Portable Document Format
OTP	One-Time Password
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
MVC	Model-View-Controller
SRS	Software Requirements Specification
SQL	Structured Query Language
JSON	JavaScript Object Notation
OOPS	Object-Oriented Programming System

RFID

Radio Frequency Identification

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